

Akshat

Real + fake audio, and GPU muscle

RTX 4060

Stage 2 PoC - 1-2 Sept

SIH26104 - Team Sonix

Your mission: Get the model the real-world voices it has never heard. Everything in Lane 1 waits on you.

THE SITUATION - READ THIS FIRST

Where we are. SONIX listens to a call in 4-second chunks and shows green / amber / red. On the academic benchmark it is strong: **1.49% EER** on ASVspoof-2019 LA. But on our own real recordings it wrongly flags roughly **half of genuine voices as fake**.

Why. The model only ever learned from clean studio recordings of real people - same good microphone, same quiet room. Our phone and laptop recordings sound nothing like that. So the model thinks: "this is not like any real voice I know, therefore fake." **The model is not broken. It is narrow.** It has never heard normal life.

The fix - three things. (1) **Show it more kinds of real:** add genuine recordings from many phones, mics, rooms and languages to training. (2) **Make it ignore silence:** quiet gaps are what break the "minimal voice" clips. (3) **Reset the alarm levels:** our amber/red numbers came from the clean test, so we re-set them using real recordings.

HARD RULE: we never train on In-the-Wild or on any eval set. Those are our honest test. Training on them would destroy the one number that proves we generalise.

THE PLAN - THREE LANES RUNNING AT ONCE

LANE 1 - FIX THE MODEL

Silence gate, more real audio, RawBoost augmentation, retrain, recalibrate. This is what stops the false alarms.

Yugal - Akshat - Yukti

LANE 2 - HONEST NUMBERS

2021-DF EER, In-the-Wild EER, codec 2x2, metrics + calibration. These are the numbers we quote to the jury.

Navya - Anubhav - Yukti

LANE 3 - A DEMO THAT CANNOT FAIL

Replay mode on validated clips, clip library, pitch, Q&A, backup video. The demo never depends on live luck.

Suryansh - Anubhav - Akshat

YOUR TASKS

P1 NOW

Collect 30-40 REAL clips

- This is the single most important task in the project tonight.
- Get teammates, friends, family to record 30-60 seconds each.
- MAXIMISE VARIETY: different phones, different laptops, noisy room, quiet room, speakerphone, earphones mic.
- Hindi and English both. Include several quiet / low-voice ones - that is the case breaking right now.
- Normal speech is fine - reading a paragraph aloud works.

Done when: 30-40 genuine clips spanning many devices and both languages.

P1 NOW

Generate 15-20 CLONED clips

- Hindi + English. Use family-emergency / 'send money now' scripts so it matches our impact story.
- For a few, play the cloned audio out of a laptop speaker and re-record it on a phone - that is what a real scam call sounds like.

Done when: 15-20 cloned clips, several of them phone-re-recorded.

P1 NOW

Keep real and fake in SEPARATE folders

- This matters a lot: it is how we get labels without any manual work.
- real\ = bonafide (label 0). fake\ = spoof (label 1). Never mix them in one folder.
- Also fill clips_manifest.csv: filename, label, language, source tool or corpus, device, notes.

```
sonix_real\real\*.wav -> all bonafide
sonix_real\fake\*.wav -> all spoof
clips_manifest.csv: filename,label,lang,source,device,notes
```

Done when: Two clean folders + a filled manifest.

P2 TONIGHT

Extract embeddings on your 4060

- Run extraction on the real and fake folders separately, then send the outputs to Yugal.
- Your GPU is one of the strongest - after this, Yugal will send you a RawBoost extraction batch.

```
python src\extract_embeddings.py --split real --audio-dir "...sonix_real\real" --
out outputs\embeddings_real --batch 8
python src\extract_embeddings.py --split fake --audio-dir "...sonix_real\fake" --
out outputs\embeddings_real --batch 8
```

Done when: outputs\embeddings_real handed to Yugal.

YOUR CHECKLIST

- ☐ 30-40 real clips, many devices
- ☐ 15-20 cloned clips
- ☐ real\ and fake\ folders kept separate
- ☐ clips_manifest.csv filled
- ☐ Embeddings extracted and sent to Yugal

Note: If you finish early, ask Yugal for the RawBoost extraction batch - your GPU is needed.

WORDS WE KEEP USING

EER	One number for how wrong the model is. Lower = better. 1.49% is strong. 50% = coin flip.
bonafide	A real human voice.
spoof	An AI-cloned / synthetic voice.
embedding	The 1024-number fingerprint of a 4-second audio chunk.
head	The small model we actually train. The big wav2vec2 front-end is frozen (never trained).
VAD	Voice Activity Detection - is anyone actually speaking in this chunk?
augmentation	Making altered copies of training audio (phone-compressed, noisy) so the model sees variety.
calibration	Making the 0-1 score mean a real probability, so thresholds are meaningful.
domain shift	Test audio looks different from training audio. THIS is our core problem.